

XIAOYING (CAMILLA) XUE

Toronto, M5G 0A6 | +1 (647) 949 9025 | camilla.xue@mail.utoronto.ca

EDUCATION

University of Toronto

Toronto, Canada

- Honours Bachelor of Science, cGPA: 3.94/4.00
 - Computer Science Specialist, Major in Statistical Science

2021 - Present

RESEARCH EXPERIENCE

Data Indexing Optimization Research Assistant

Toronto, Canada

- *Professor Niv Dayan* Fall 2025
 - Conducted comprehensive benchmarking of **RocksDB** to evaluate and identify read/write bottlenecks across different indexing methods and data configurations.
 - Designed, implemented, and assessed a **minimal perfect hash indexing** technique integrated within the SST data-block read and write pipelines, reducing costly binary search hops and achieving a measurable 5% improvement in read performance.

Client-Server Interface Design Research Assistant

Toronto, Canada

- *Professor Qizhen Zhang* Summer 2025
 - Conducted in-depth analysis of **PostgreSQL** codebase to uncover client-server communication pathways and define function semantics for systematic benchmarking of communication latency.

WORK EXPERIENCE

Software Engineer Intern

Toronto, Canada

- *Quanta Technology* Summer 2024 – Fall 2025
 - Contributed as a full-stack developer in an **Agile** environment and used **Git** for version control to develop a finite-state-control workflow portal with **React** for the front end and **C#** for the back end, significantly reducing manual overhead and improving operation accuracy.
 - Developed a web-based software tool for utility companies, effectively managing complex and deeply nested tree-structure JSON data to enable robust **CRUD** operations via **RESTful APIs**, and interactive visualization for hierarchical data, significantly enhancing data management efficiency.
 - Collaborated with cross-functional teams to design database solutions that support a smooth transition and scalable handling of over 1 million data records, implementing **PL/SQL** procedures and triggers to automate business processes and ensure data integrity.

RELEVANT EXPERIENCE

Pet Adoption Website Developer

Toronto, Canada

- *Computer Science Team Project* Fall 2023
 - Planned and designed UI/UX workflows using **HTML**, **CSS**, and **JavaScript**, implemented a dynamic front-end with **React** to deliver a responsive user experience.
 - Developed **RESTful** web services using **Django REST Framework**, extending back-end functionality to support CRUD operations across multiple user roles with custom **Python** code.
 - Performed API quality assurance using **Postman** to test request methods, input validation, and access permissions, ensuring robust and secure back-end interactions.

TEACHING & OUTREACH

Teaching Assistant

Toronto, Canada

- STA130: An Introduction to Statistical Reasoning and Data Science Fall 2025, Winter 2024
 - Hosted weekly 2-hour tutorials for a group of 25 students, assisted students' research projects.

Volunteer Highschool Mentor

Toronto, Canada

- WICS (Women in Computer Science) Fall 2023
 - Held regular one-on-one mentoring sessions with high school students to provide academic support.

SKILLS

- **Programming Languages:** C, C++, C#, Java, TypeScript, JavaScript, Python, R, SQL, HTML, CSS
- **Frameworks & Libraries:** React, Bootstrap, Django, NumPy, Pandas
- **Languages:** English, Mandarin, Cantonese, Japanese